PROOF OF CONCEPT · TELECOM INDUSTRY

Turning Churn Data Into a Retention Strategy

A predictive machine-learning approach to identify, explain, and act on customer churn risk — built on 100,000 customer records and grounded in Latin American market research.

Presented By

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GCI World 2026 · Report

Agenda: What We'll Cover

Our structured approach to solving *churn* in Latin American telecom markets.

01
Market Context
Why churn is the #1 strategic cost in Latin American telecom.

02
Business Problem
Shifting from reactive damage control to proactive retention.

03
Data Insights
Four hypotheses tested against 100k customer profiles.

04
ML Model
CatBoost performance, metrics, and tuning logic.

05
The Solution
From risk scores to an operational CRM decision layer.

06
Impact
Quantified financial gain and strategic recommendations.

How is Latin America Today: The High Cost of Customer Mobility

Saturation and high mobility make customer retention a survival imperative in LatAm.

20-40%

Annual Churn

Annual churn rate among regional telecom operators.

5-10x

Higher cost

More expensive to acquire than to retain a customer.

Failure-Driven

Technical

Churn problems are mostly focused on service failures and outages.

3%

Revenue Drop

Recent decline in operational revenue in regional markets like Peru.

16x

Prediction Savings

Predicting churn is sixteen times less costly than new customer acquisition efforts.

+85%

Potential profit boost from a 5% churn reduction.

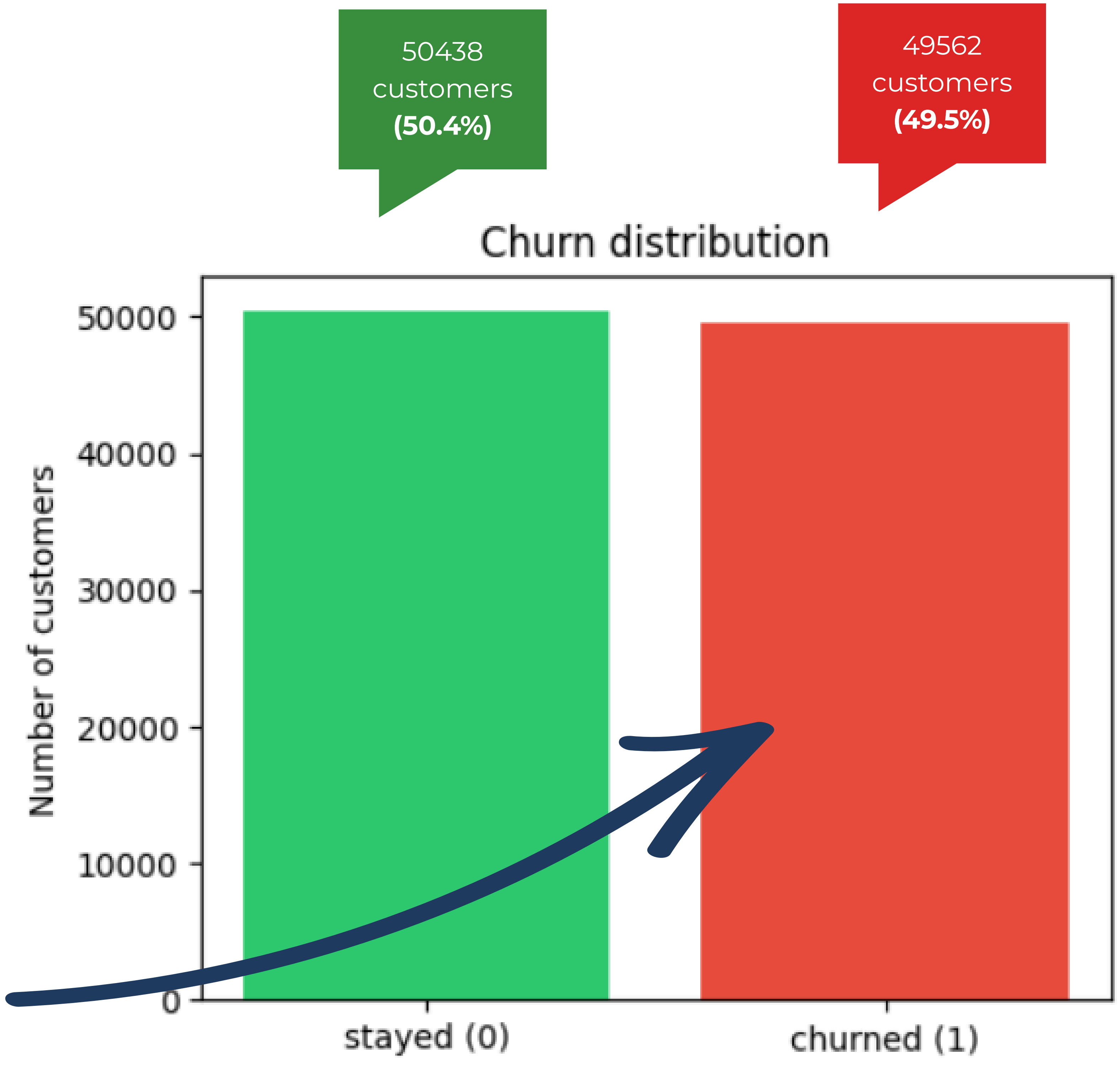
- Abarca Sánchez, Y., Barreto Rivera, U., Barreto Jara, O., & Díaz Ugarte, J. L. (2022). "Customer loyalty and retention at a leading telecommunications company in Peru." Revista Venezolana de Gerencia, 27(98), 729-743.
- Arteaga Ventura, C., Rosales Meza, X. A., & Ponce Álvarez, S. (2024). "Lean service and BPM to improve service levels in a telecommunications company." III International Congress of Industrial Engineering 2024, University of Lima, Peru.
- Correa, A. (2019). "Consumer choice and local network effects in the Colombian mobile telecommunications market." Journal of Quantitative Methods for Economics and Business, 28, 160-182.
- Mocoroa, J. M. (2025). "The Challenges of the Argentine Universal Service in Telecommunications: New Regulations, Old Problems." Digital Journal of Administrative Law, Externado University of Colombia, No. 33, 77-108.
- Pérez Rave, J. I. (2010). "Opportunities for improvement, from standpoint workers, in a customer service line for mobile telephony." Ingeniare. Chilean Journal of Engineering, 18(1), 84-94.
- Saha, L., Tripathy, H. K., Gaber, T., El-Gohary, H., & El-kenawy, E.-S. M. (2023). "Deep Churn Prediction Method for Telecommunication Industry." Sustainability, 15(5), 4543.

My Business Problem: Data Without Insight

- ✓ Current retention is reactive: cancellation requests arrive too late.
- ✓ 49.6% **churn** rate in historical data reveals structural issues.**(9.6% above the LATAM average)**
- ✓ Need to shift from "Who left?" to "Who is about to **leave**?"

KEY QUESTION

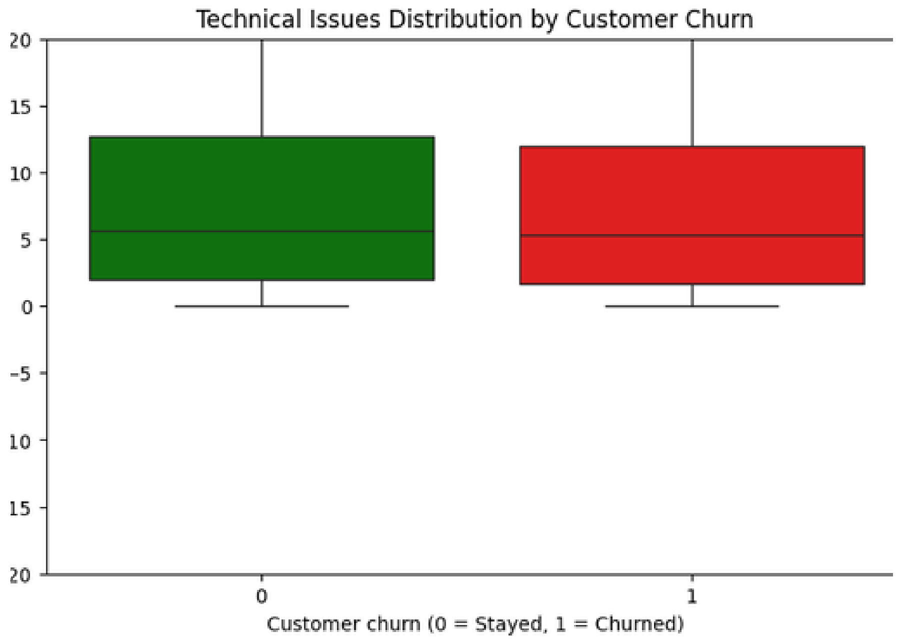
They are **churning**, but why? And how do I retain them? What is the best offer to keep them?

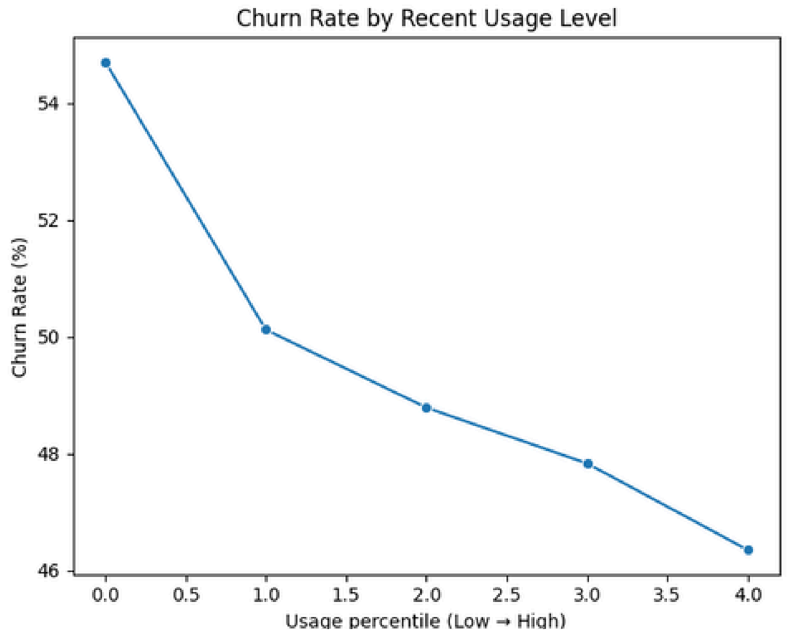


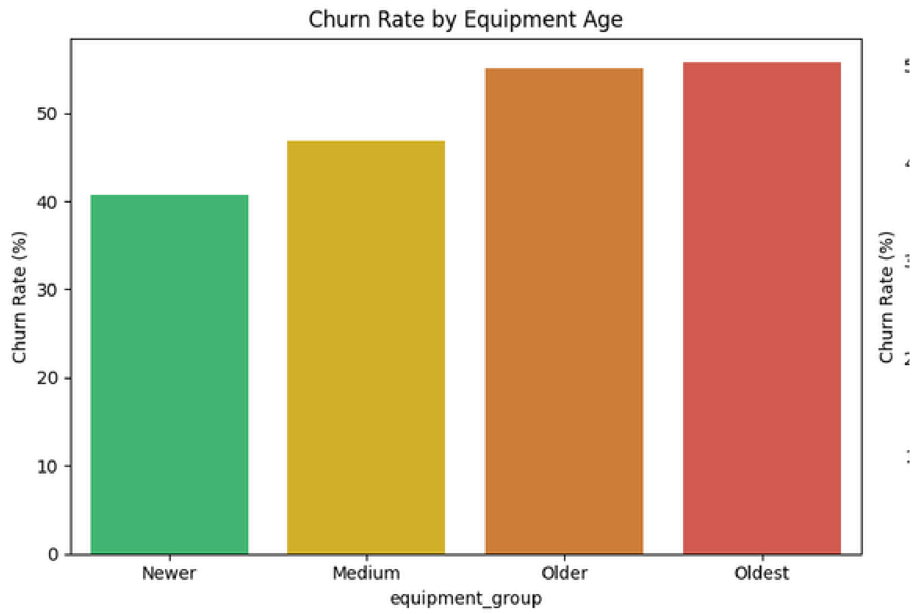
*These data are based on a sample of 100,000 people; the corresponding charts and related data can be found in the notebook.

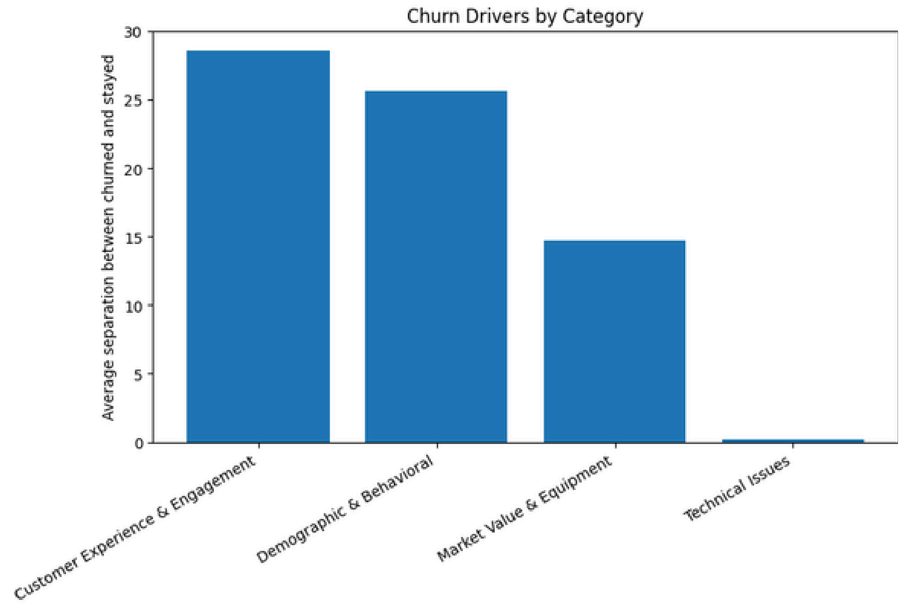
Tested Hypotheses: What Drives Flow?

Key factors analyzed to understand and predict customer **churn** behavior.









H1: Technical Reliability

Are failures or poor signal quality driving drop-offs?

Limited Impact

H2: Engagement Decay

Does disengagement precede the cancellation request?

Confirmed

H3: Device Obsolescence

Does handset age correlate with brand abandonment?

Confirmed

H4: Behavior & Engagement Trends

Is behavior and engagement stronger signal than Technical Uses?

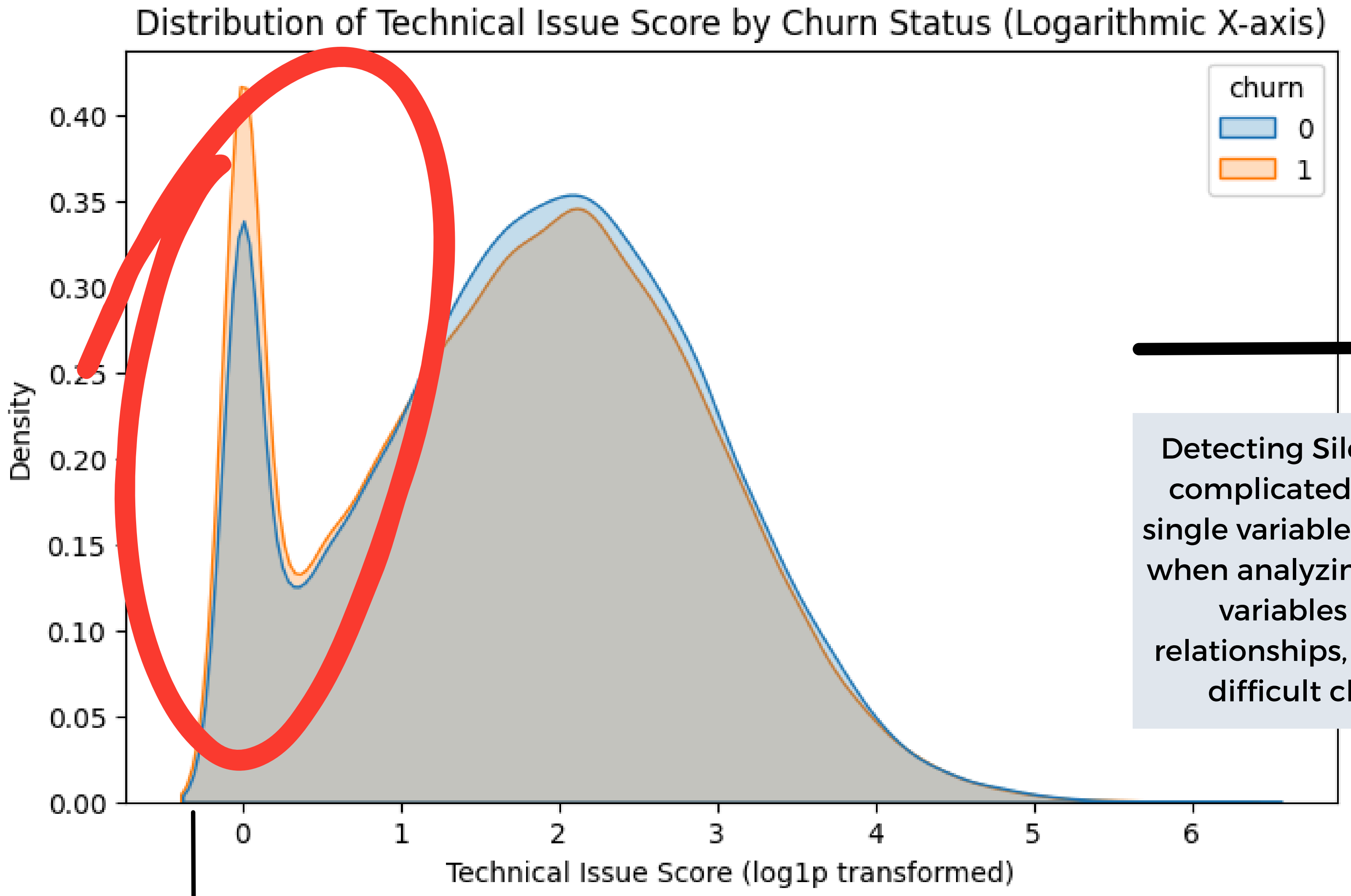
Strongly Confirmed

Churn is a multidimensional facto = Experience/Engagement [High] + Tech Issues [Low]

1. Hypotheses are based on market research and problems stated in the references on slide #1.
2. Results obtained from the EDA; please refer to the notebook for more detailed information
3.The data are based on a sample of 100,000 people; the corresponding charts and related data can be found in the notebook.

Insight : Silent churn

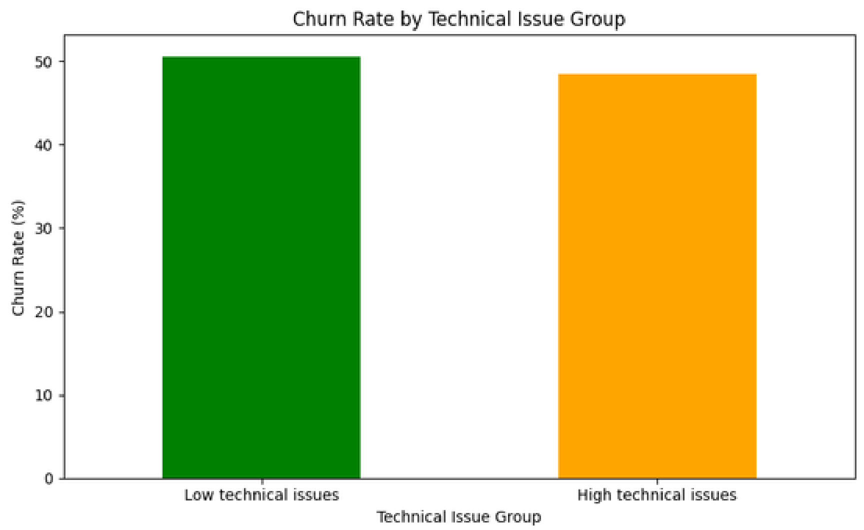
Technical reliability is a baseline expectation, not a differentiator. Customers with "clean" service churn as much as those with issues.



Low Technical Issues & High Churn??? = Silent Churn

```
df["technical_issue_score"] = (
    df["drop_vce_Mean"] +
    df["drop_dat_Mean"] +
    df["blck_vce_Mean"] +
    df["blck_dat_Mean"]
```

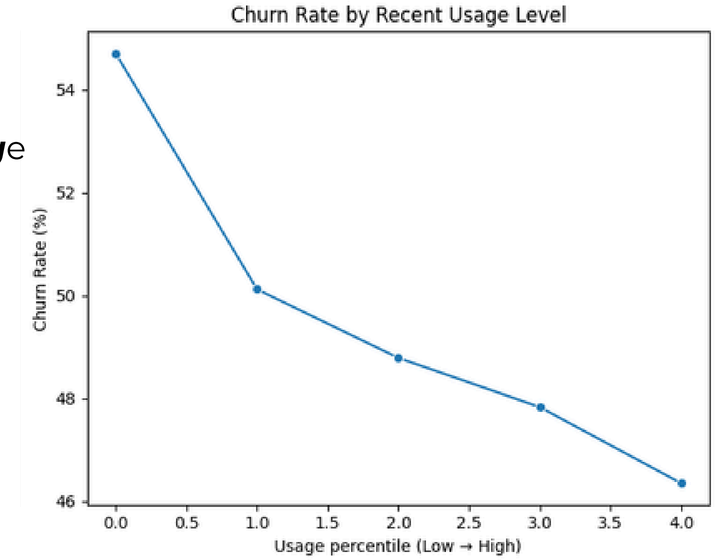
Detecting Silent churn is complicated with just a single variable; in fact, even when analyzing a group of variables or their relationships, it remains a difficult challenge



50.6%
Churn rate for customers with **MINIMAL** issues.

48.5%
Churn rate for customers with **RECURRENT** issues.

""BUT a ***systematic decay in service usage*** can serve as an early warning signal to detect this silent churn".



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Solution: ML+ Operational CRM Layer

¿How do I know who is about to leave?

Through a machine learning AND classification model that predicts the churn probability for each customer and defines the priority.

¿How do I know why they are leaving?*

By integrating an Explainability Layer (SHAP values) that identifies the main risk factors triggering the alert.

¿How do I Know how I retain them?*

Through a Prioritization Matrix that crosses Churn Risk with Customer Value (totmrc_Mean), automatically assigning a personalized retention strategy.

1. Churn Prediction Model (ML Layer)

Input: Customer features

Output:
- Churn probability
- Predicted churn classification

↓

2. Risk Segmentation Layer

Segments predicted churners according to churn probability:

High Risk: 80%-100% churn probability **Medium Risk:**50%-80% churn probability **Low Risk:**Threshold-50% churn probability

↓

3. Customer Value Assessment

Uses totmrc_Mean as the customer value indicator.

Low Value: totmrc_Mean < P50 **Medium Value**: P50 ≤ totmrc_Mean < P75 **High Value**: totmrc_Mean ≥ P75

↓

4. Retention Prioritization

Combines: **Churn Risk + Customer Value**

Generates retention priority:

Priority 1: High Risk + High Value **Priority 2**: High Risk + Medium Value **Priority 3:** Medium Risk + High Value **Priority 4**: Low Risk / Low Value

↓

5. Explainability Layer

Identifies why the customer is considered at risk Using **SHAP** feature contributions

Generates risk drivers:

- Customer Engagement Risk - Value Perception Risk -Customer Relationship Risk - Technical Reliability Risk

↓

6. Retention Action Strategy

Defines the recommended action:

- Retention offer
- Customer contact
- Loyalty benefit
- Follow-up strategy

*Risk drivers are identified through statistical patterns (SHAP values) rather than human causality, and the recommended actions are strictly suggestive. These business dimensions and action strategies are not learned by the Machine Learning model during training, nor are they a direct output of the algorithm; they represent an external operational framework subject to individual protocols and assumption (.please refer to the data table notebook)

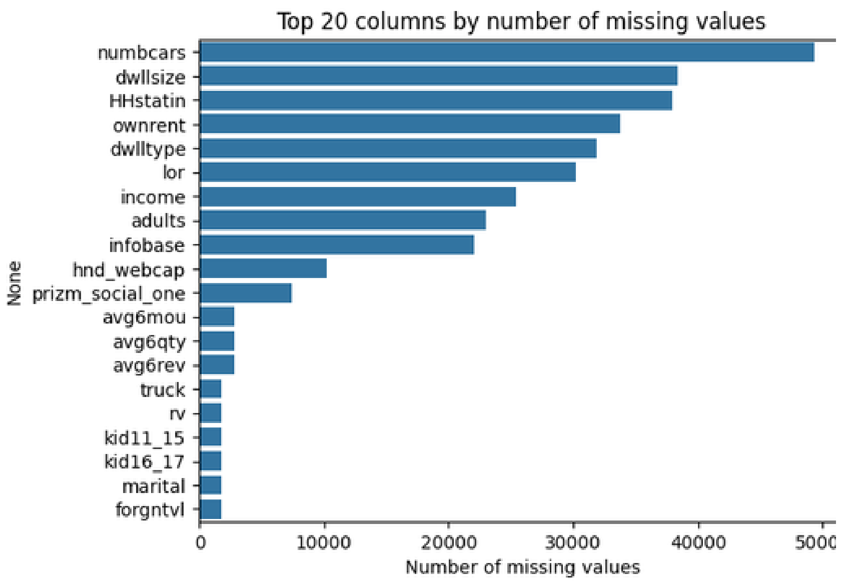
How it Works?: The Machine Learning Engine

The Brain: Why CatBoost?

Native Handling of Categorical and Missing Data

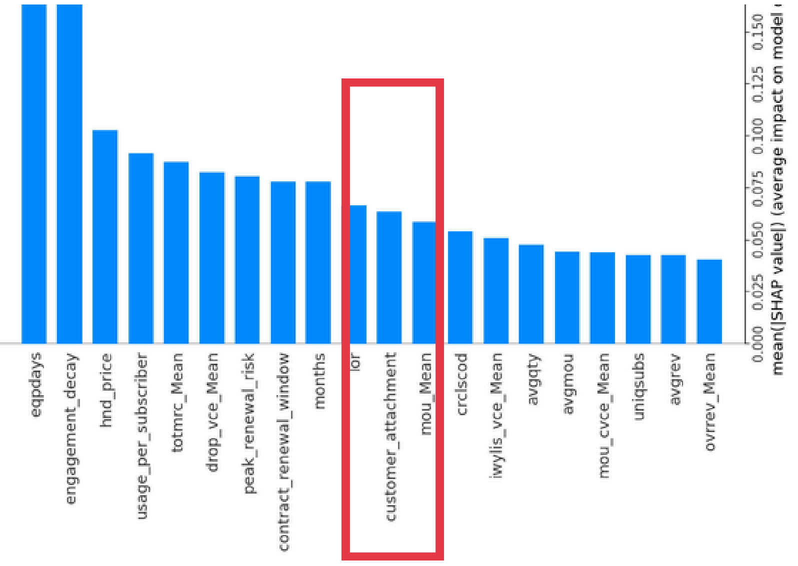
The dataset contains **21 categorical variables** and a **substantial number of missing values.**

CatBoost can effectively use demographic and customer information, even when some records are incomplete, while maintaining high prediction accuracy.



Complex Non-Linear Relationships among Features

Customer churn in telecommunications is driven by complex interactions rather than isolated variables. **CatBoost effectively captures COMPLEX relationships identified during the EDA,** such as customer_attachment, a feature that integrates Months in Service, Unique Subscriptions, and Active Subscriptions. **By analyzing these factors together, the model can identify stronger churn signals than when each variable is considered individually**



Superior Predictive Performance (Baseline Comparison)

During the initial evaluation phase, CatBoost demonstrated consistent superiority over XGBoost across almost all critical metrics:

ROC-AUC: 0.6892 (vs. XGBoost: 0.6895)

Recall: 0.6481 (vs. XGBoost: 0.6429)

F1-score: 0.6376 (vs. XGBoost: 0.6349)

CatBoost supports explainability techniques such as Feature Importance and SHAP analysis, which are essential for the Retention Intelligence Layer.

How We Did It? : The Machine Learning Engine

Data Cleaning & Feature Engineering

Dimensionality Reduction: Removed non-predictive variables to streamline the dataset.

Missing Data & Matrix Management: Handled a complex matrix of 21 categorical features and high missingness (particularly in demographics) using median and mode (most frequent value) imputation.

Advanced Features:
Engineered 9 interaction variables to capture complex customer behaviors flagged as critical during the Exploratory Data Analysis (EDA).

K**ey Insights: AUC Boost & Behavioral Shift**
AUC Improvement: New Interaction features successfully captured complex behavioral patterns missed by the original variables.
The Model's "Brain" Evolved:
Before: Feature importance was dominated by pure tenure (months). After: The newly engineered engagement_decay variable jumped to #1 in importance.

Hyperparameter Tuning & Threshold

Randomized Search: A ParameterSampler was utilized to test random combinations of hyperparameters, including tree depth (depth), learning rate (learning_rate), number of iterations (iterations), and L2 regularization (l2_leaf_reg).

The Outcome: The optimal combination was selected (e.g., learning_rate: 0.1, l2_leaf_reg: 3, iterations: 500, depth: 5), maximizing the validation ROC-AUC

Business-Driven Choice: Performed a rigorous threshold sensitivity analysis to align the model's classification cutoff with core business priorities and financial metrics (the economics).

Key Insights: Asymmetric Costs & Threshold Optimization

- Asymmetric Error Costs: Model errors do not carry the same weight:
  - False Negative (Missed Churner): Costs $269.94 (the total lost customer value over 6 months).
  - False Positive (Accidental Offer): Costs only $19.78 (the baseline campaign intervention cost).
  - **The Strategy: Since losing a customer is 13.6 times more expensive than wasting an offer, a highly aggressive retention approach is required.**
-
- Moving Beyond the Standard 0.50 Threshold:
  - At 0.50 Threshold: The model is balanced mathematically but leaves too many customers behind (Recall is only ~66%).
  - At 0.30 Threshold: While Precision drops to 53.8%, Recall skyrockets to 94.4%.
  - ***Business Impact: It is far more profitable to scale up retention offers—even if some are unnecessary—to guarantee capturing 9 out of 10 real churners.***

- ***Best Practices: Ensuring Robustness & Avoiding Overfitting***
- ***Strict Data Splitting (Hold-out & Validation): Split the 100,000-record dataset into independent subsets: 70% Training, 15% Validation, and 15% Test.***
- ***The unseen final test set confirmed a stable ROC-AUC of 0.693, closely matching training results and proving no overfitting.***
- ***Early Stopping: Applied early_stopping_rounds=50 during CatBoost training. Training stopped optimally at iteration 476, discarding subsequent versions that were simply fitting to noise.***
- ***L2 Regularization & Complexity Control:***
- ***Feature Selection & Streamlining: Removed redundant variables using a correlation matrix to combat the curse of dimensionality.***
- ***Conservative Economic Validation (P10 Scenario):***
- ***Locked the 0.30 decision threshold using the 10th Percentile (P10) of profitable scenarios to mitigate risk and safeguard against overoptimistic projections.***
- ***Ensures that the projected USD 91,677.36 net value remains realistic and achievable under shifting market conditions.***

What We Achieved ? : The Machine Learning Engine

0.693

AUC

The system achieves a **69% probability of correctly distinguishing a high-risk customer** from a loyal one, staying well above the 50% random baseline

0.94

Recall

Captures **9 out of 10** actual churners before they leave.

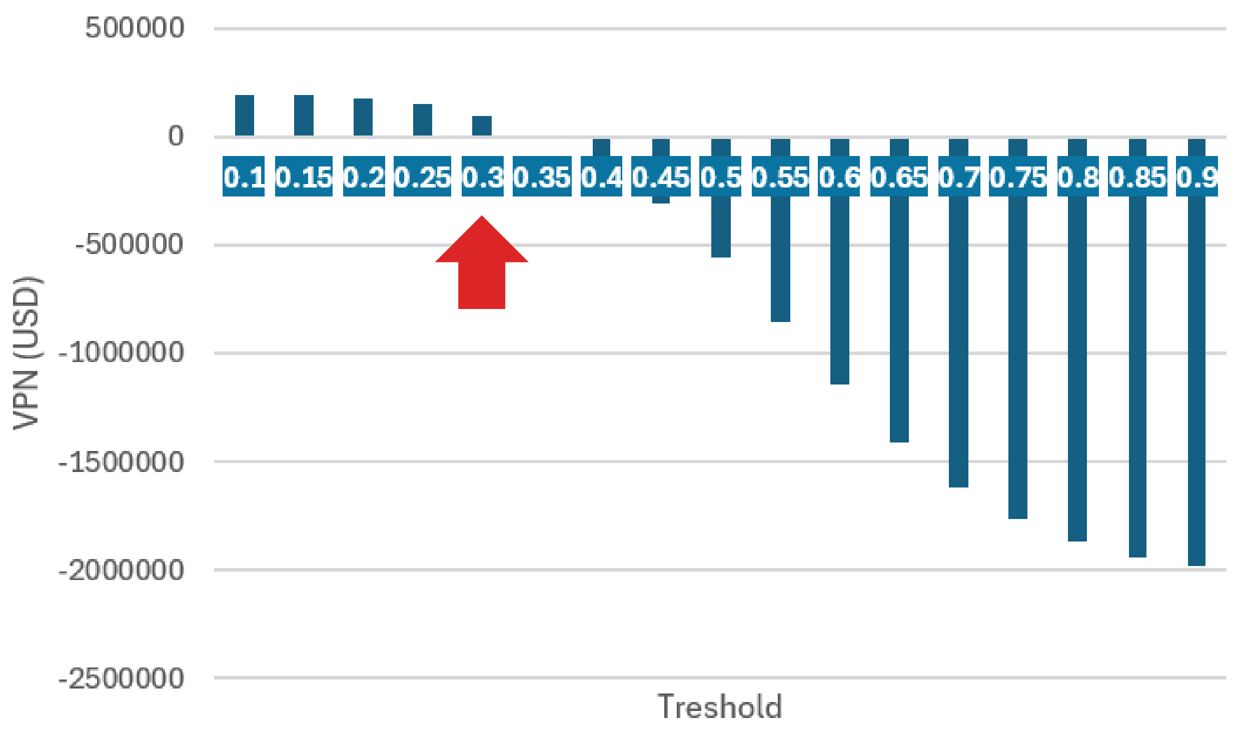
What is our priority?

We prioritize r**evenue protection through aggressive detection (94% Recall)**, under the premise that it is far better to invest **USD 19.78* in** a preventive action than to lose a customer with a lifetime value of hundreds of dollars.(Losing a customer is **13.6 times more expensive** than wasting an offer)

¿The Economics?

+ $91,677

This amount represents **the protected revenue** from the customers we successfully saved, minus campaign costs and model error costs (both undetected customers and offers sent by mistake).



Economic Optimization: The system balances revenue saved against misclassification costs via a strict net return profit formula:

$$\text{Net Value} = (\text{TP} \times r \times v) - (\text{FP} \times c) - (\text{FN} \times v)$$

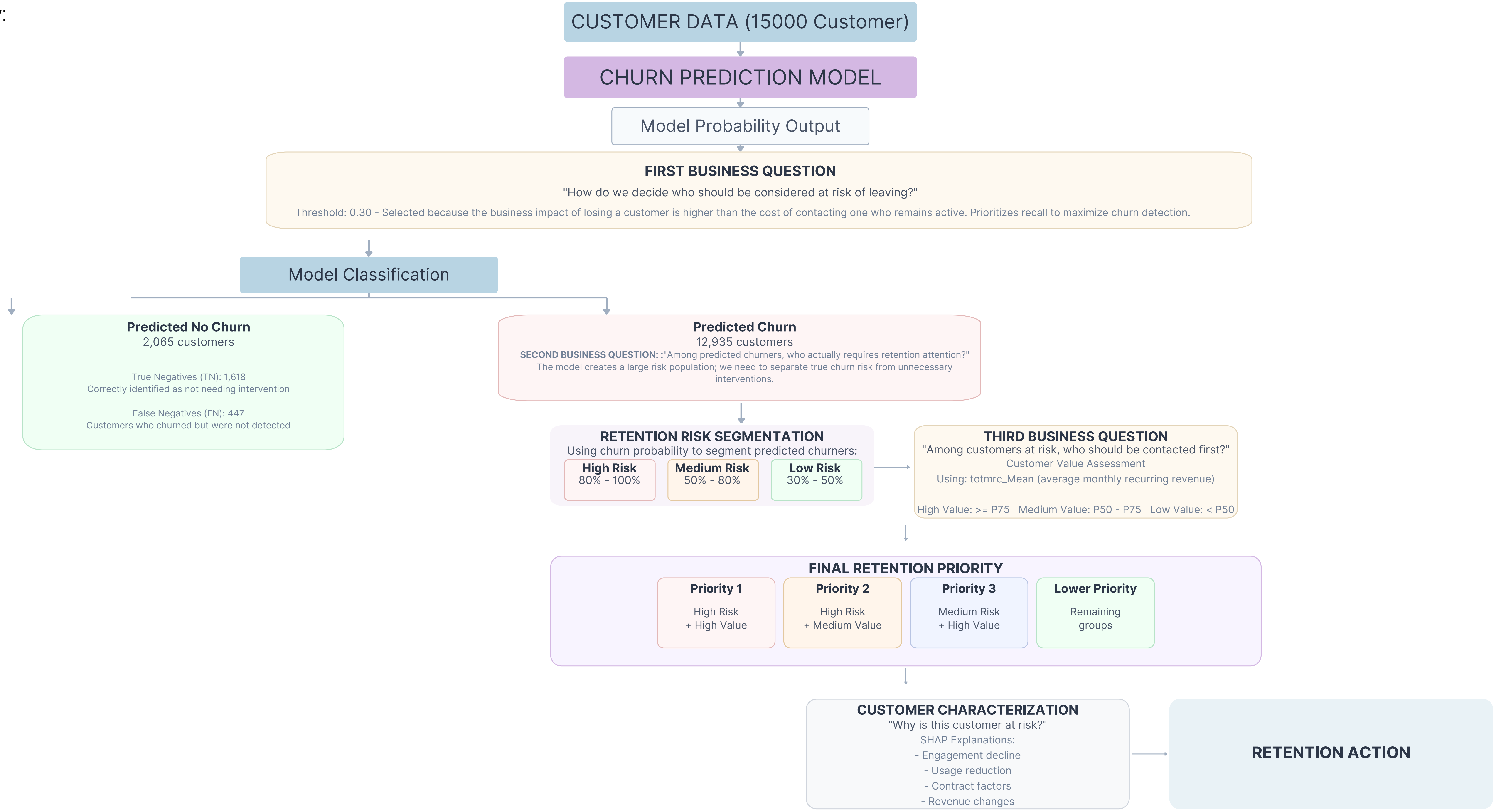
Financial Framework Summary

- Campaign Costs & Success (c, r): Intervention cost is set at USD 19.78 per user. The retention success rate is assumed at 17% based on industry literature (Silva Vílchez, 2024).
- Customer Lifetime Value (v): Calculated stable revenue over a conservative 6-month horizon using the baseline monthly recurring revenue metric (totmrc_Mean).
- Cost of Model Errors: * False Negatives (FN): Missed churners leading directly to lost revenue.
- False Positives (FP): Loyal customers targeted by mistake, causing wasted operational campaign spend.
- Conservative Selection (P10): Instead of chasing the absolute highest theoretical profit, a 10th Percentile (P10) threshold criteria was applied to guarantee a realistic, risk-mitigated business expectation.(threshold = 0.3
- TP (True Positives): Customers at real risk of churning who were correctly identified by the model.

- Value refered by the paper : Abarca Sánchez, Y., Barreto Rivera, U., Barreto Jara, O., & Díaz Ugarte, J. L. (2022). "Customer loyalty and retention at a leading telecommunications company in Peru." Revista Venezolana de Gerencia, 27(98), 729-743.
- For a detailed breakdown of Economics' calculations and assumptions, please refer to the "Threshold Sensitivity Analysis" chapter in the notebook.
- Result obtained with a data of 15.000 customers

From Prediction to Profit: ML Model + Business Intelligence Layer

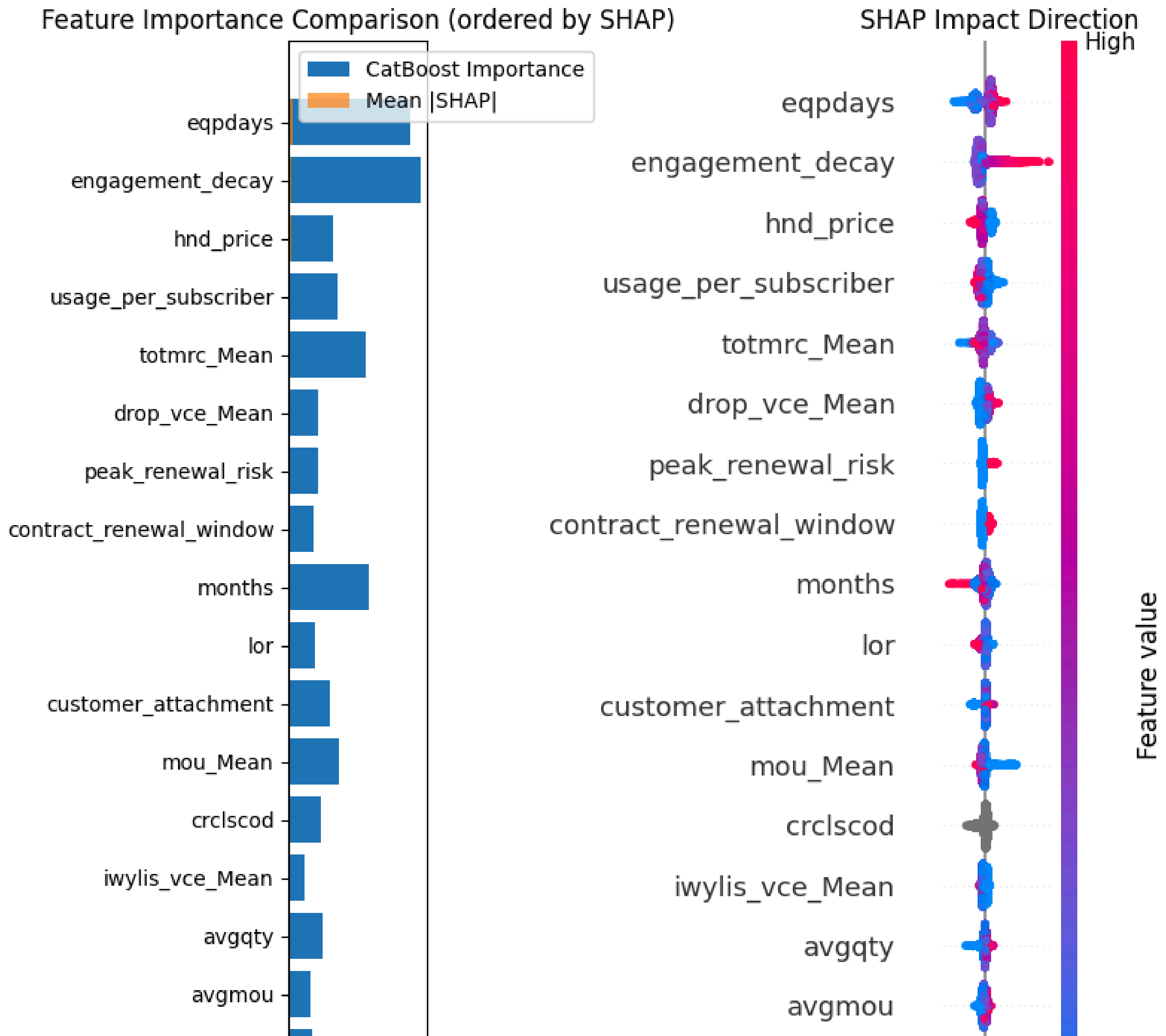
Model Workflow:



From Prediction to Profit: CUSTOMER CHARACTERIZATION & RETENTION

¿How do I know why they are leaving?*

By integrating an Explainability Layer (SHAP values) that identifies the main risk factors triggering the alert.



¿What is SHAP?

SHAP acts as the model's "translator," clearly explaining the exact customer behaviors and reasons behind why a specific user is flagged at risk of churning.

¿Why SHAP?

Local & Global Explainability: SHAP breaks down exactly how much each individual variable pushes a specific customer toward churning, while simultaneously mapping overall company-wide trends.
Consistent Feature Importance: Unlike traditional importance metrics that can be easily distorted, SHAP calculates the true mathematical contribution of each feature across all possible combinations.

```
====================================
Customer position: 13768
====================================
              feature  shap_value
95   engagement_decay    0.253652
11      drop_vce_Mean    0.104518
5         ovrrev_Mean    0.079710
64   prizm_social_one    0.048496
6         vceovr_Mean    0.044789
15      unan_vce_Mean    0.044191
91           creditcd    0.044176
4         ovrmou_Mean    0.042770
77            marital    0.034613
57             avgqty    0.033268
75                lor    0.032765
55             avgrev    0.028410
51           asl_flag    0.026919
54             totrev    0.024668
3             da_Mean    0.023662
```

```
==========================================
Customer position: 11842
==========================================
                   feature  shap_value
95        engagement_decay    0.324978
1                 mou_Mean    0.237577
97    usage_per_subscriber    0.222015
92                 eqpdays    0.210482
68               hnd_price    0.167995
9               change_mou    0.147894
2              totmrc_Mean    0.085303
10              change_rev    0.075051
59                 avg3qty    0.069708
61                 avg6mou    0.067957
33           peak_vce_Mean    0.065613
39           mou_opkv_Mean    0.061359
31         iwylis_vce_Mean    0.055460
30        mouowylisv_Mean    0.050832
56                  avgmou    0.039197
```

From Prediction to Profit: CUSTOMER CHARACTERIZATION & RETENTION

¿How SHAP Tells Us Why Customers Are Leaving?

SHAP Importance Extraction

Capture and rank all variables based on their true mathematical impact on the customer’s churn risk, isolating the exact forces pushing them to leave.

Actionability Filtering:

Filter out demographic or biased features, keeping only actionable signals that marketing or sales can influence: behavior, value, technical performance, and relationship status.

Risk Dimension Grouping & Scoring

Categorize the remaining features into the four predefined business dimensions (Technical, Engagement, Value, and Relationship).

Maximum Score Assignment

Calculate the total SHAP contribution for each dimension and automatically assign the customer to the single highest-scoring risk pillar to trigger their tailored retention offer.

From Prediction to Profit: CUSTOMER CHARACTERIZATION & RETENTION

The model reduces:
***Manual analysis of large customer populations.**
***ineffective campaigns targeting customers with low retention potential.**
***delayed reactions after customers already decided to leave.**

	retention_priority	churn_probability_x	customer_value	main_churn_driver_category	recommended_action	top_churn_drivers
0	Priority 1	0.956707	70.0000	Customer Engagement Risk	Customer engagement campaign focused on increa...	[{'feature': 'totmrc_Mean', 'feature_value': 1...
1	Priority 1	0.941960	69.9900	Customer Engagement Risk	Customer engagement campaign focused on increa...	[{'feature': 'mou_opkd_Mean', 'feature_value':...
2	Priority 1	0.923529	59.9900	Customer Engagement Risk	Customer engagement campaign focused on increa...	[{'feature': 'engagement_decay', 'feature_valu...
3	Priority 1	0.922548	104.9700	Value Perception Risk	Commercial retention offer focused on improvin...	[{'feature': 'engagement_decay', 'feature_valu...
4	Priority 1	0.922256	59.9900	Value Perception Risk	Commercial retention offer focused on improvin...	[{'feature': 'hnd_price', 'feature_value': 29....
5	Priority 1	0.921715	59.9900	Value Perception Risk	Commercial retention offer focused on improvin...	[{'feature': 'eqpdays', 'feature_value': 617.0...
6	Priority 1	0.920097	59.9900	Value Perception Risk	Commercial retention offer focused on improvin...	[{'feature': 'drop_vce_Mean', 'feature_value':...
7	Priority 1	0.919432	59.9800	Customer Engagement Risk	Customer engagement campaign focused on increa...	[{'feature': 'engagement_decay', 'feature_valu...
8	Priority 1	0.919271	59.9900	Customer Engagement Risk	Customer engagement campaign focused on increa...	[{'feature': 'hnd_price', 'feature_value': 29....
9	Priority 1	0.919253	77.1975	Customer Relationship Risk	Customer lifecycle intervention focused on str...	[{'feature': 'peak_renewal_risk', 'feature_val...

The solution enables:
better campaign success rates.
data-driven decision making.
Solution Framework

list of references

1. Abarca Sánchez, Y., Barreto Rivera, U., Barreto Jara, O., & Díaz Ugarte, J. L. (2022). Fidelización y retención de clientes en una empresa líder de telecomunicaciones en Perú. Revista Venezolana de Gerencia, 27(98), 729–743.
2. Arteaga Ventura, C., Rosales Meza, X. A., & Ponce Álvarez, S. (2024). Lean service y BPM para aumentar el nivel de servicio en una empresa del sector de las telecomunicaciones [Trabajo de investigación]. III Congreso Internacional de Ingeniería Industrial 2024, Universidad de Lima, Perú.
3. Correa, A. (2019). Elección del consumidor y efectos de red locales en el mercado de telecomunicaciones móviles de Colombia. Revista de Métodos Cuantitativos para la Economía y la Empresa, 28, 160–182.
4. Kupera. (2026). Metodología para formulación de proyectos / Design Thinking Methodology: Frameworks and user experience design (Documento metodológico interno). Laboratorios de Sueños.
5. Mocoroa, J. M. (2025). Desafíos del servicio universal argentino en materia de telecomunicaciones: nuevos reglamentos, viejos problemas. Revista Digital de Derecho Administrativo, Universidad Externado de Colombia, (33), 77–108.
6. Pérez Rave, J. I. (2010). Oportunidades de mejoramiento de una línea de atención al cliente de telefonía móvil desde la perspectiva del asesor. Ingeniare. Revista chilena de ingeniería, 18(1), 84-94.
7. Saha, L., Tripathy, H. K., Gaber, T., El-Gohary, H., & El-Kenawy, E.-S. M. (2023). Deep Churn Prediction Method for Telecommunication Industry. Sustainability, 15(5), 4543.
8. Silva Vílchez, Y. K. (2024). Elaboración y ejecución de un plan de marketing para la retención de clientes que quieran dar de baja su línea de celular de la unidad de negocio de servicios [Trabajo de suficiencia profesional, Pontificia Universidad Católica del Perú]. Repositorio Institucional PUCP.
9. Herramientas de IA Utilizadas para la Investigación
10. Perplexity: Disponible en https://www.perplexity.ai/search/estudios-de-mercado-sobre-las-Tn2vnzsmR3yXqlHsGuyQmA.
11. NotebookLM: Disponible en https://notebooklm.google.com/notebook/ff710dc7-ce3c-44ee-8162-545bf8ac994b.